

Recovery Theorem and the Risk Aversion. Evidence from the Czech Republic

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Abstract

The main goal of this study is to obtain expectations regarding the future development of the exchange rate using derived option probabilities and based on them to calculate the perception of risk by investors. More specifically, this study deals with the application of the Ross recovery theorem in the FOREX market using the CZK/EUR exchange rate. From a theoretical point of view, I offer an expression of the Ross recovery theorem using different Numeraire and I also apply a novel approach to the calculation of the implied risk premium. The results show that both subjective and risk-neutral densities are not unbiased probability estimates of a future exchange rate. However, the results show a significant increase in the implied risk aversion during the Covid19 pandemic.

D4, G22, E32

Risk Aversion, Ross Recovery Theorem, Risk-neutral densities, Subjective densities, Risk premium, Covid19, FX options, Czech Republic, Czech Crown.

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1 Introduction

Financial markets can provide useful information about the perception of risk by individual market participants. Options contracts are particularly useful tools, where the reason for their usefulness is due to the fact that one of the key factors determining the prices of these derivative contracts is the probability that certain events will occur. However, these contracts also include information on investors' risk aversion, as investors demand a premium for their uncertainty.

The approach presented in this study makes it possible to derive the implied risk-neutral and subjective densities of exchange rate developments using option prices. The obtained densities can theoretically allow us to extract market beliefs regarding the future. This is because subjective probability, in this case, refers to the market's estimate of the true probability density, i.e., the market's (subjective) expectation of returns, volatility, etc... It should be noted that the risk-neutral probabilities are different from subjective (and real-world) densities. Hypothetically, if the investors were risk-neutral, the risk-neutral probabilities denote what the expectations about future developments would need to be so that the present prices were identical to those observed.

Regarding subjective densities, for a long time, it was considered to be an unsolvable problem to separate risk aversion and the subjective probability that we obtain from the prices of derivative contracts from each other. However, a novel solution was offered by Ross (2015), who uses specific restrictions on the functional form of the utility function of a representative investor together with other limitations, to get the desired solution. Due to the knowledge of these subjective densities, we can focus, for example, on the development of the risk premium over time and whether there is an improvement in predictive ability, i.e., whether these obtained densities reflect real market developments.

This study presents an application of the Ross recovery theorem in the FOREX market. Based on his work, I extract the subjective densities of individual participants from previously acquired risk-neutral densities. Furthermore, I propose a novel approach using a combination of these results to obtain a risk premium. Finally, I compare statistics constructed from the obtained densities and investigate their predictive abilities.

1.1 Literature Review

The use of the Breeden and Litzenberger (1978) method for estimating risk-neutral density has become especially popular for determining market expectations. For example, the work of Jackwerth and Rubinstein (1996) studies the S&P 500 index, and the authors are interested in the predictive power and probability of tail events implied by the obtained densities. Alonso (2005) then applies the same analysis to the Spanish IBER35 index. Furthermore, the work of Figlewski (2008) deals with the impact of monetary policy decisions of the Fed on the development of the S&P 500 index through an event study regarding risk-neutral density quantiles. Other publications focus on subjective probabilities and try to extract them. These are papers by Sahalia and Lo (2000,) and Jackwerth (200), Liu et al. (2007), and Rosenberg et al.

(2001), where all these works assume that the subjective density distribution is equal to the realized historical distribution. The opposite approach is characterized by the follow-up work of Bliss and Panigirtzoglou (2004), where the authors use risk-neutral densities to derive the risk aversion coefficient and test the predictive power. Most similar to this study is Hanke et al. (2020), which deals with quantiles of risk-neutral densities of the world stock markets and their response to the COVID-19 pandemic.

Application of the Breeden and Litzenberger method to non-equity markets is less common, mainly due to the lower number of option quotations. In the case of currency markets, the most well-known publication is Malz (1997) and Malz (2014), where the author uses additional assumptions to interpolate the volatility surface. The first study by Malz (1997) assumes a second-order polynomial form of implied volatility, whereas the second work uses an interpolation by the clamped spline method.

It is also possible to list other papers that all use the Malz methodology in different periods and on other currencies. For example, the study of Cuaresma et al. (2014) deals with risk-neutral densities in the case of Central European countries. Similar analyses are provided by the Turkish Central Bank’s working paper Korkmaz et al. (2019) and Bank of New Zealand working papers, Gereben (2002) and Lewis (2012). There is also the work of Babua (2008) on this topic regarding Polish zloty and the euro. Finally, Castren (2005) examines the risk-neutral density of the exchange rates of the three new EU member states, Poland, Czech Republic, and Hungary.

The literature on the extraction of subjective densities is quite limited, although the Ross Recovery Theorem has attracted some other authors who have built upon Ross’s research ever since his publication. Regarding the theoretical foundations, Ross’s work was followed by Car and Yu (2012), who have shown that the recovery theorem can be extended to the continuous state-space setting. Another paper by Qin and Linetsky (2017) shows how to express Ross recovery in the more general state-space of Borel right processes. Backwell (2015) formalized the Recovery Theorem and added the extension, where he relaxed the assumption of state-independent preferences. In contrast, Dubynskiy and Goldstein (2013) and Borovicka et al. (2016) present a critique of the recovery theorem, which focuses on the decomposition of the stochastic discount factor.

Empirical work literature applying the recovery theorem is also relatively sparse, and the results vary considerably. Martin et al. (2019) applies the recovery theorem to analyze long-dated bonds and the yield curve movements. Audrino (2014) examines the S&P 500 option prices to investigate whether the recovery yields predictive information and found that market timing strategies based on recovered moments significantly outperform their counterparts based on their risk-neutral densities. Bakshi et al. (2018) apply the recovery theorem using Treasury bond futures and reject the recovery theorem. Jackwerth and Menner (2020) show that recovered physical distributions based on the S&P 500 index fail to predict future returns. These disappointing results are even more potent when authors add, as they claim, economically reasonable constraints. Lan and Li (2018) builds on the method of Backwell (2015) using

an improved strategy for combining option data with the Recovery Theorem to estimate the subjective densities; authors then investigated whether the Recovery Theorem yields predictive information and rejected the hypothesis.

On the other hand, Kiriū and Hibiki (2019) propose a robust method for the Ross Recovery Theorem that uses prior information and ensures numerical stability. Gagnon et al. (2019) investigate whether moments recovered using recovery theorem perform better to forecast realized volatility. The authors found evidence that these factors significantly improved realized volatility forecasts. Flint and Mare (2018) implement the Ross Recovery Theorem using option prices of the stock market in South Africa. Sanford (2018) shows that more robust results can be achieved using a multivariate Markov chain. Appel and van Mare (2019) follow up on their work with specific functional forms of pricing kernel and simulated data for determining the robustness of the given method.

2 Options and Risk neutral densities

Options are contracts that allow one party to sell/buy an asset at a pre-agreed price in the future. Since options are quoted for several different strike prices at the same maturity, they allow observing the market's perceived probability distribution.

The basic model for determining an FX option's price is the generalized Black Scholes model. (Garman and Kohlhagen, 1978) This model assumes a lognormal movement of the underlying asset and constant volatility over time. As an input for determining the price using this model, six parameters are used $S_t, \tau, X, r_t, q_t, \sigma(t, X)$. These are the spot price, the time to maturity, the exercise price, and the risk-free domestic and foreign interest rate. The last necessary parameter is the volatility $\sigma(t, X)$. However, the value of this parameter does not correspond to the observed historical volatility of the underlying asset. Therefore, the reverse engineering method is used to determine the value of implied volatility. Knowledge of implied volatility makes it possible to assess how market participants perceive the market risks associated with individual options. For this reason, it is typical for the options market that prices are quoted in implied volatility terms.

The currency options market is further specific because there is only a limited number of options with different strike prices. However, the standard of most markets is the existence of at least three options with different exercise prices at the same maturity. In the Czech environment, three quotations of European options are used. These are at the money (atm_t), i.e., delta neutral call/put option, 25 delta put option, 25 delta call option.

The issue is more complicated by the fact that instead of quoting the three options, related so-called option strategies are quoted instead. The most often quoted as Risk reversal (rr_t) and butterfly strategy (bf_t). Risk reversal (rr_t) is an options strategy where the investor simultaneously buys an out-of-the-money call option and sells an out-of-the-money put option. Butterfly (bf_t) is a strategy that usually consists of 4 options. The portfolio is created by buying a call option with a relatively low strike price, buying a call option with a relatively high strike price, and

selling two call options with a strike price that is midway between these two strike prices. The relationship between individual quotations is expressed as follows (Reiswich et al., 2010):

$$\sigma_t^{call}(\Delta) = atm_t + bf_t(\Delta) + rr_t(\Delta)/2 \quad (1)$$

$$\sigma_t^{put}(\Delta) = atm_t + bf_t(\Delta) - rr_t(\Delta)/2 \quad (2)$$

The approach presented in Breeden and Litzenberger (1978) is used in this study to determine the risk-neutral density. The approach is based on the simple idea that price always expresses the expected discounted return on a given asset. In the case of a European call option, its price is expressed as follows:

$$C(X) = e^{-rt} E^q \max(S_t - X, 0) = e^{-rt} \int_X^\infty (x - X) f_q dx \quad (3)$$

E^q denotes the conditional expectation at time t according to the risk-neutral probability measure q , S_t denotes the price of the asset at maturity, X denotes the strike price, e^{-rt} represents the discount factor and f_q is the risk-neutral density. Assuming continuity and differentiability, a probability density can be obtained using the second derivative with respect to the strike price.

$$f_q = e^{rt} \frac{\partial^2 C(X)}{\partial X^2} \quad (4)$$

3 Ross recovery theorem

Let us now focus on the decomposition of the risk-neutral density, which includes both the real expectation of future movements and the risk premium. For a better understanding, we would ideally be interested in the subjective probability density f_p of investors; although it occurs in the valuation, it is modified through the pricing kernel, denoted by m_{t+1} . In the simple tree model, the pricing kernel expresses the ratio of marginal utilities of a representative investor Lucas (1978). Using simple algebraic manipulations, it can be shown that only if the ratio m_t is equal to one, thus the investors are risk-neutral, and then the option price would reflect subjective market expectation. (Formally E^p denotes the conditional expectation at time t according to the probability measure p .)

$$C(X) = e^{-rt} \int_X^\infty (x - X) \frac{f_q}{f_p} f_p dx = E^p(m_{t+1} \max(S_t - X, 0)) \quad (5)$$

Under strong assumptions, the subjective probabilities of market participants can be extracted from the risk-neutral densities as is noted by Ross (2015). The first assumption of this method is the existence of a discrete and bounded state space Θ , consisting of θ_i , where $i \in (1, 2, \dots, n)$, the other assumptions are the absence of arbitrage and a complete market. The next assumption is that the previously defined pricing kernel is m_t transition independent, and thus there exists a representative investor with a time additive inter-temporal utility. Provided that all conditions are met, the subjective probabilities can be obtained by the Ross recovery theorem.

The solution of the Ross recovery theorem can be shown in the following way. Assume a representative investor who solves the following two-period inter-temporal optimization problem given that that investor is in a generic θ_i initial state of the world (Backwell, 2015):

$$\max u(c_{t,i}, c_{t+1,i}) = u(c_{t,i}) + \delta E^p u(c_{t+1}) = c_{t,i} + \delta \sum_{j=1}^n u(c_{t+1,i,j}) f_{i,j} \quad (6)$$

Given a budget constraint:

$$c_{t,i} + \sum_{j=1}^n c_{t+1,i,j} p_{i,j} = w \quad (7)$$

$u(c_{t,i}, c_{t+1,i})$ denotes the two-period utility function. $f_{i,j}$ denotes the real probability of transition from the generic initial state θ_i to the generic final state θ_j . $p_{i,j}$ represents the price of the Arrow Debreu security contingent on the initial state θ_i , which pays 1 unit of Numeraire¹ if the next state is θ_j , and zero otherwise. $c_{t,i}$ denotes initial consumption at time t and final consumption at time $t+1$ contingent on the future state of the world is then denoted by $c_{t+1,i,j}$, δ denotes a discount factor and w is positive constant representing the wealth. The first order conditions then lead to the following solution for the price of state-contingent claim for every state i, j

$$p_{i,j} = \delta \frac{u(c_{t+1,i,j})'}{u(c_{t,i})'} f_{i,j} = m_{i,j,t+1} f_{i,j} \quad (8)$$

Finally, according to Backwell (2015), we assume an equilibrium condition on consumption, that ensures that the consumption is assumed to be dictated only by the state. In other words, for every i, j , the following holds:

$$c_{t,j} = c_{t+1,j} = c_i \quad (9)$$

The above optimal solution for all i, j can be further simplified using the equilibrium condition of time homogeneity of consumption. It should be noted that this significant simplification and the assumption of time homogeneity is one of the main weaknesses of the Recovery theorem. This is due to the fact that it rules out some useful pricing kernel properties, such as possible path-dependence in expectations and different transition state prices in different periods, i.e., we assume that all time periods have identical structures for prices and probabilities. Therefore, the pricing kernel is the transition independent, which means that it is independent of any calendar time.

$$p_{i,j} = \delta \frac{u(c_j)'}{u(c_i)'} f_{i,j} = m_{i,j} f_{i,j} \quad (10)$$

Since this applies to all i, j in θ , the result can be rewritten in matrix form as:

$$P = \delta D^{-1} F D \quad (11)$$

P is a $n \times n$ matrix of Arrow Debreu prices, F is a $n \times n$ matrix of subjective probabilities of investors and D is an $n \times n$ diagonal matrix of marginal utilities, which consists of d_i on

¹Unit of account, for formal treatment see: Geman et. al (1995)

the diagonal and zero outside the diagonal. Using matrix algebra and the knowledge that the matrix F is a stochastic matrix, i.e., the sum of individual rows must be one or $F1 = 1$, where 1 is the vector of the ones. Due to the Perron Frobenius theorem, which guarantees the existence of a unique real eigenvalue, the problem is reduced to finding the largest eigenvalue and its corresponding eigenvector. The calculation is as follows, where we defined $D^{-1}1 = z$.

$$Pz = \delta z \quad (12)$$

4 Stochastic Discount Factor in FOREX market

So far, we have dealt with the general valuation of a risky financial instrument. Let us now turn to the FOREX market and the risk premium associated with it. We begin with a definition of the gross return of a risky asset denominated in foreign currency as in Backus et al. (2001):

$$1 = E^p(m_{i,t+1}(1 + r_{r,t+1})) \quad (13)$$

Next, we consider an asset denominated in foreign currency. We adopt the same approach as before for pricing this foreign-denominated asset. By $*$ we denote foreign variables, i.e., $m_{i,j}^*$ denotes foreign pricing kernel and $r_{r,t+1}^*$ foreign gross return:

$$1 = E^p(m_{i,t+1}^*(1 + r_{r,t+1}^*)) \quad (14)$$

By using the no-arbitrage argument we could alternatively convert foreign instrument returns into domestic returns and value them using domestic pricing kernel, where FX_t denotes the exchange rate at a time t and $E^p FX_{t+1}$ denotes the conditional expectation at a time t of the exchange rate at a time $t + 1$.

$$1 = E^p(m_{i,t+1}(1 + r_{r,t+1}^*) \frac{E^p FX_{t+1}}{FX_t}) \quad (15)$$

This leads to the following equality between pricing kernels and exchange rate development:

$$m_{i,t+1} \frac{E^p FX_{t+1}}{FX_t} = m_{i,t+1}^* \quad (16)$$

This reason, as was stated before, is that if equality is violated and both currencies are traded, there is an arbitrage opportunity, as there would exist expected excess return by converting to another currency. In conclusion, using the no-arbitrage argument, we have derived the equivalence of these two procedures for the valuation of foreign assets. Combining this result with the Ross recovery theorem stated above gives us the following relation:

$$p_{i,j} = m_{i,t+1} \frac{E^p FX_{t+1}}{FX_t} f_{i,j} = m_{i,t+1}^* f_{i,j} \quad (17)$$

By combining this result with the assumption that the exchange rate development can be described by different Markov transition matrix², we obtain an alternative definition of the Ross recovery theorem.

$$p_{i,j} = \delta \frac{u(c_j)'}{u(c_i)'} f_{i,j} f x_{i,j} \quad (18)$$

I use $f x_{i,j}$ to denote the ratio of the exchange rate (FX in matrix form). This can be rewritten in matrix form using the fact, that the product of two stochastic matrices is a stochastic matrix, i.e., $F(FX) = (FX)F = Q$ and the Perron-Frobenius theorem thus can be applied:

$$P = \delta D^{-1}(FX)ED = \delta D^{-1}QD \quad (19)$$

5 Methodology

To determine the price of the option for non-quoted values, it is necessary to use interpolation and extrapolation methods, which lead to the need to make certain assumptions about the form of implied volatility. The calculation method used in this study is the Vanna Volga method from Castagna and Mercurio (2007). Vanna Volga method is based on the assumption of dynamic hedging and the creation of a locally risk-free portfolio. Using the knowledge of implied volatility, we have defined all possible option prices at different exercise prices, which is also equivalent to the knowledge of risk-neutral probability. (Blake and Rule, 2015)

More formally, Castagna and Mercurio (2007) assume that a risk-free portfolio can be created consisting of a long position in a call option, a short position in the underlying instrument, and three short positions in other options. Due to this, the portfolio is not only risk-free in the delta but also Vega, Vanna, and Volga hedged, i.e., the portfolio then is hedged not only against shifts in the underlying asset up to the second order but also against movements in implied volatility also up to second order. In other words, there is a perfect local hedge, provided we have a sufficient number of other options. The great advantage of the Vanna Volga method is its relative simplicity. Therefore, unlike other methods such as SABR, it can be easily applied to less liquid markets with fewer quotations.

Even more formally, let's assume three different options available, each with different strike prices denoted by X_i , and x_i representing the weight of the given option in the portfolio. The portfolio has the following form:

$$\pi = C(X) - \Delta X - \sum_{i=1}^3 x_i C(X_i) \quad (20)$$

Using Ito's lemma, we get the following expression for portfolio development:

$$d\pi = dC(X) - \Delta dX - \sum_{i=1}^3 x_i dC(X_i) \quad (21)$$

²Formally, we must also assume the same condition as in the case of consumption regarding equilibrium value.

To erase the stochastic parts, it is necessary to determine the individual weights (x_i) and the size of an underlying asset (Δ). Using the well-known relations between the Greeks, we obtain the following conditions for Vega, Vanna, and Volga:

$$v = \frac{\partial C(X)}{\partial \sigma_t} = \sum_{i=1}^3 x_i \frac{\partial C(X_i)}{\partial \sigma_t} = \sum_{i=1}^3 x_i v_i \quad (22)$$

$$vd2 = \frac{\partial^2 C(X)}{\partial \sigma_t^2} = \sum_{i=1}^3 x_i \frac{\partial C(X_i)}{\partial \sigma_t} \frac{d2_i}{X_i \sigma_t \sqrt{t}} = \sum_{i=1}^3 x_i v_i d2_i \quad (23)$$

$$vd2d1 = \frac{\partial^2 C(X)}{\partial \sigma \partial X} = \sum_{i=1}^3 x_i \frac{\partial C(X)}{\partial \sigma_t} \frac{d2_i d1_i}{\sigma_t} = \sum_{i=1}^3 x_i v_i d2_i d1_i \quad (24)$$

Vega (v) denotes a partial derivative with respect to the change in implied volatility of the underlying currency pair. Vanna and Volga are expressed using Vega. This is a system of three equations with three unknowns which leads to the following weights for Vanna Volga's secured portfolio:

$$x_1 = \frac{v \ln(X_2/X) \ln(X_3/X)}{v_1 \ln(X_2/X_1) \ln(X_3/X_1)} \quad (25)$$

$$x_2 = \frac{v \ln(X/X_1) \ln(X_3/X)}{v_2 \ln(X_2/X_1) \ln(X_3/X_2)} \quad (26)$$

$$x_3 = \frac{v \ln(X/X_1) \ln(X/X_2)}{v_3 \ln(X_3/X_1) \ln(X_3/X_2)} \quad (27)$$

To determine the size of the delta, it is necessary that the portfolio is delta-secured. (Since this approach is only an approximation, we assume constant volatility.) The result is the following position in the underlying asset:

$$\Delta = \frac{\partial C(X)}{\partial X} - \sum_{i=1}^3 x_i \frac{\partial C(X_i)}{\partial X_i} \quad (28)$$

The result is a locally risk-free portfolio, the price of which is the sum of the delta hedge and Vanna Volga hedge. To determine the state prices based on the approach presented in Breeden and Litzenberger (1978), we can use the second derivative of the portfolio price with respect to the strike price.

The main practical problem of the empirical implementation of the Ross recovery theorem is, therefore, the estimation of the matrix P containing the Arrow Debreu prices $p_{i,j}$. Ross (2015) proposes a solution using the current state prices vector s_t . Using these prices, we obtain the state space, which consists of n realization prices and m maturities. However, this is not the required matrix P . The difference is that the vector s_t consists of prices (prices of the contingent claims), which give one unit of Numeraire if state j occurs due to the fact that we are in state i . In contrast, the matrix P determines the prices of Arrow Debreu securities for every state i and j . However, there is one row of the matrix P , which is already known. It is the vector s_t

that corresponds to the transition time given by the transition matrix P .

For each period, I am able to obtain state prices s_t using the (Breedon and Litzenberger, 1978) method with different maturities up to time t . Using the assumption of time homogeneity of matrix P , I obtain the following relation between state prices and matrix P , which can be written in vector form as follows:

$$s_t = \frac{\partial^2 C(X)}{\partial X^2} \quad (29)$$

$$s_t = s_{t+1}P \quad (30)$$

Due to the assumption of time homogeneity, the relation can be aggregated for all maturities as follows: $A = BP$ under the condition $P > 0$. Where $A = (s_2, \dots, s_m)$ is a matrix $n \times m$ and $B = (s_1, \dots, s_{m-1})$ is a matrix $n \times m$, provided that $n \leq m$. Due to the numerical instability of the matrix P , the method of generalized ridge regression with a prior probability matrix $\tilde{P}$ is used. The estimation of the matrix P can, therefore, be written as the following optimization problem ³ as in Kiriou and Hibiki (2019):

$$P = \operatorname{argmin}_{P>0} (AP - B)^t (AP - B) + \lambda (P - \tilde{P})^t (P - \tilde{P}) \quad (31)$$

$$P = \operatorname{argmin}_{P>0} y^{fit}(\lambda) + \lambda y^{reg}(\lambda) \quad (32)$$

$\tilde{P}$ is a $n \times n$ matrix containing a prior information. The matrix form is based on the assumptions that $p_{i,j}$ is similar to $p_{i+k,j+k}$ for all $k < \min(n-i, n-j)$. This basically guarantees the unimodality of transition probabilities within this transition matrix and thus the highest values are located on the diagonal of the matrix. This formulation effectively restricts the state space over which probabilities must be estimated, which is necessary given limited data. On the other hand, it also eliminates states that are large outliers, this also means disaster events that are the main reason for people to buy put options. For this reason, it can significantly distort distributions with long tails or significant skewness. This means that if we are in state i at time t , we are most likely to move near it. In general, the matrix $\tilde{P}$ has the following form:

$$\tilde{P} = i_0 \begin{pmatrix} \sum_{k=1}^{i_0} s_{k,1} & \dots & s_{n,1} & 0 & 0 \\ \sum_{k=1}^2 s_{k,1} & \dots & s_{n-1,1} & s_{n,1} & 0 \\ s_{i_1,1} & s_{i_1,1} & \dots & s_{i_1,n-1,1} & s_{i_0,n} \\ 0 & s_{1,1} & s_{2,1} & \dots & \sum_{k=n-1}^n s_{k,1} \\ 0 & 0 & s_{1,1} & \dots & \sum_{k=i_0}^n s_{k,1} \end{pmatrix} \quad (33)$$

For simplicity, the estimation is performed with the assumption that there is an odd number of states θ_i and the middle column of the matrix $\tilde{P}$ is denoted $i_0 = (n+1)/2$. It should be

³Practically the optimization problem can be rewritten as n generalized ridge regressions. The non-negative least squares algorithm is then used for the calculation itself due to the need to maintain positive transition probabilities.

noted here, that the values are defined as spot moneyness (X/S). The optimal lambda is then determined through problem-specific function $h(\lambda)$:

$$\operatorname{argmin} h(\lambda) = \frac{y^{fit}(\lambda) - y^{fit}(0)}{y^{reg}(\infty) - y^{reg}(0)} + \frac{y^{reg}(\lambda) - y^{reg}(\infty)}{y^{reg}(0) - y^{reg}(\infty)} \quad (34)$$

y^{fit} denotes residuals and y^{reg} denotes the difference between the estimated and the prior matrix. In the denominator, there is the maximum possible range, and in the numerator, there is a range for the given value of λ . Kiri and Hibiki (2019) showed λ monotonically increases y^{reg} and monotonically decreases y^{fit} .

The actual implementation was done in the following way. The implied state price surface obtained above is discretized in $m = 54$ evenly spaced maturities from 1 month to one year, which roughly approximates the weekly maturities. This is due to the fact that if we used only six non-overlapping periods (with different lengths) we would obtain only six transitions which allow for at most six states. This would result in inaccurate measurement due to the coarse grid. Concerning the strike dimension, we create a moneyness grid from 0.85 to 1.25 ($n = 29$), which seems to be sufficient given the high persistence of the exchange rate. The moneyness grid size is also approximately equal to the half size of Jackwerth and Menner (2020), which deals with much more volatile stock prices. Finally, the state price matrix S consisting of $(s, \dots, s_m)$ is then split into two submatrices with a lag of 3 months, meaning that the recovered physical transition probabilities represent expectations for 3 months ahead.

6 Empirical estimation

6.1 Data

The Refinitiv database was used as a data source. Bid and ask option end of the day quotations from the Butterfly strategy, Risk reversal strategy and at the money option with maturities of 1,2,3,6,9 and 12 months on each business day since the beginning of the year 2018 were obtained.⁴ 25 delta put option, 25 delta call option quotations are derived as in equations 1 and 2. Strike prices from the OTC quotations are computed as in Reiswich et al. (2010). The strike price for at the money delta is computed as follows:

$$X = F_t e^{\frac{1}{2}\sigma_t^2 t} \quad (35)$$

Strike price for other values of (unadjusted spot) delta is obtained as follows:

$$X = F_t e^{\frac{1}{2}\sigma_t^2 t + -\sigma_t \sqrt{t} N^{-1}(e^{qt} \Delta_s)} \quad (36)$$

⁴Specifically following tickers were used: EURCZK1MO=, EUCZ1MRR=, EUCZ1MBF=, EURCZK2MO=, EUCZ2MRR=, EUCZ2MBF=, EURCZK3MO=, EUCZ3MRR=, EUCZ3MBF=, EURCZK6MO=, EUCZ6MRR=, EUCZ6MBF=, EURCZK9MO=, EUCZ9MRR=, EUCZ9MBF=, EURCZK1YO=, EUCZ1YRR=, EUCZ1YBF=.

The last day obtained is October 7, 2020. The resulting value was determined as the average of the current bid and ask prices. IBOR quotes with the corresponding maturity were used as a proxy for the risk-free rate. The mid-price quotation was used for the spot rate. For overall statistics on the FX market, the Czech National Bank survey data were used.

6.2 Risk Neutral Densities

The study empirically deals with the foreign exchange market in the Czech Republic, where it focuses mainly on the period after the end of the foreign exchange commitment, emphasizing the period of the COVID-19 pandemic. To begin the description of the foreign exchange market in the Czech Republic, the following Figure 1 shows the structure of the Czech foreign exchange market from domestic banks' perspective. It can be seen that over 90% of foreign exchange transactions with the Czech koruna carried out by domestic banks are with the Euro and the US dollar. It can be observed that forward, and FX swaps occupy the largest share in notional volume, and there is a much smaller share of options.

Figure 1: FX transactions of Czech banks (daily volume)

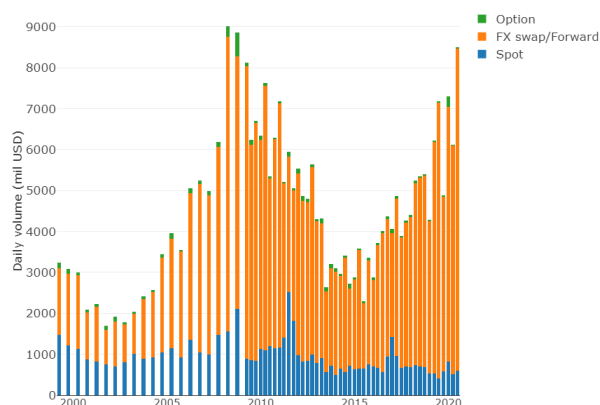
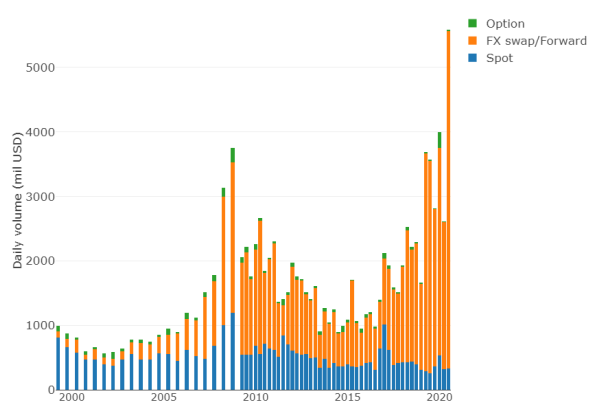


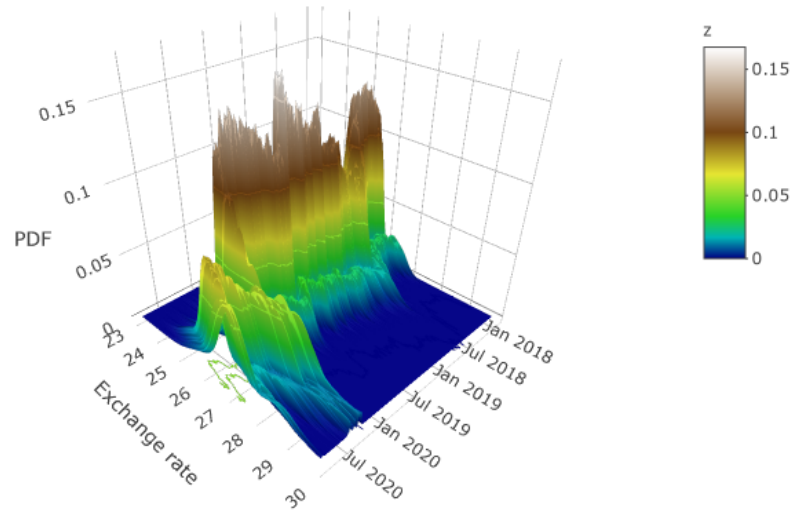
Figure 2: CZK/EUR statistics



Source: Czech National Bank

I obtained the risk-neutral density using the Breeden and Litzenberger method. Let us focus on the time series of the risk-neutral probability density of the three-month CZK/EUR exchange rate from 2018 to 2020. From its development, as is shown in Figure 3, we can see a stable and almost symmetrical probability distribution. The mean and median of the given distribution oscillates between 25 and 26 CZK/EUR. Furthermore, the 10% and 90% quantiles of the implied risk-neutral distribution consistently fall within the range of 25 to 27 CZK/EUR, with the distance between individual quantiles remaining approximately constant until 2020, or even slightly decreasing in specific periods.

Figure 3: Risk-neutral density



Source: own calculation, Refinitiv

However, already in the first quarter of 2020, we have observed a significant worldwide increase in the uncertainty associated with the COVID-19 pandemic. The Czech koruna weakened to the level of 28 CZK/EUR, and there was a significant change in the implied density of the CZK/EUR exchange rate. This means that the obtained implied densities showed an increase in the mean of the distribution and an increase in the standard deviation, interquartile range, and Pearson and Kelly skewness, as shown in Figure 4. These statistical moments peaked in the first half of 2020. In particular, the month of April indicates the highest increase in uncertainty about exchange rate developments, which is manifested by the highest standard deviation in the observed period.

Another statistical moment examined is skewness, which is shown in Figure 5. The COVID-19 pandemic results show an increase in skewness, indicating a greater probability of depreciation of the Czech koruna. However, the results probably include a significant increase in investors' risk aversion, as will be illustrated later. This can be intuitively explained by the investor's fear of further depreciation of the Czech Koruna, which can be explained by the Czech Republic's perception as an emerging market and CEE country.

Figure 4: RND: Standard Deviation and Skewness

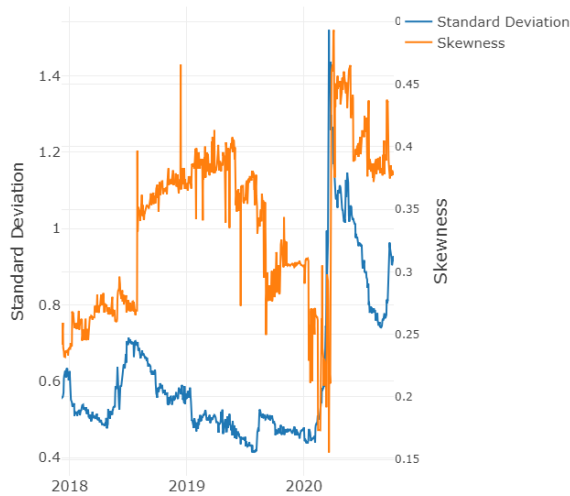
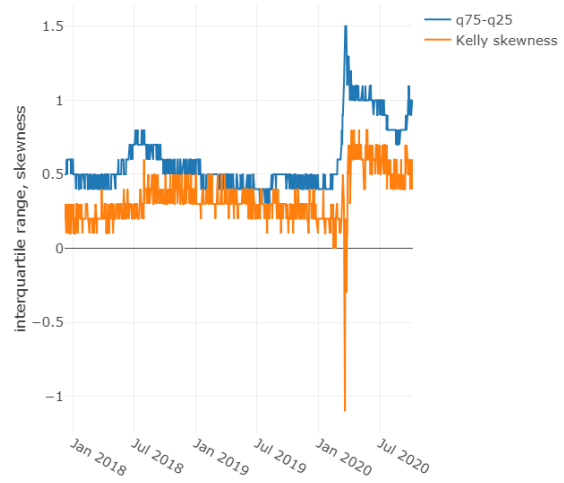


Figure 5: RND: Kelly Skewness and interquartile range



Source: own calculation, Refinitiv

Next, let's focus on the predictive power of derivative contracts, namely whether the market was able to predict the COVID-19 pandemic. Firstly, we can observe that the increase in the variance and individual quantiles of the risk-neutral density occurred essentially simultaneously with the sudden movement in the spot exchange rate, as is shown in Figure 6. However, this can also be interpreted as a sign that the market did not expect such a sharp movement, and due to this, the reaction of investors was probably even more pronounced. Since May, the situation has gradually calmed down, but the risk-neutral density remains highly dispersed and skewed to the right.

I also tested the predictive ability of the entire risk-neutral density by using the probability integral transform (Diebold, 1998). The results indicate limited predictive ability; the values on the left side of the histogram (Figure 7) prevail, which means a more frequent depreciation of the Czech koruna than expected. This was formally confirmed by the Kolmogorov-Smirnov statistic, where the null hypothesis about the uniformity of the obtained distribution was rejected.

Figure 6: Risk-neutral density at the time of settlement

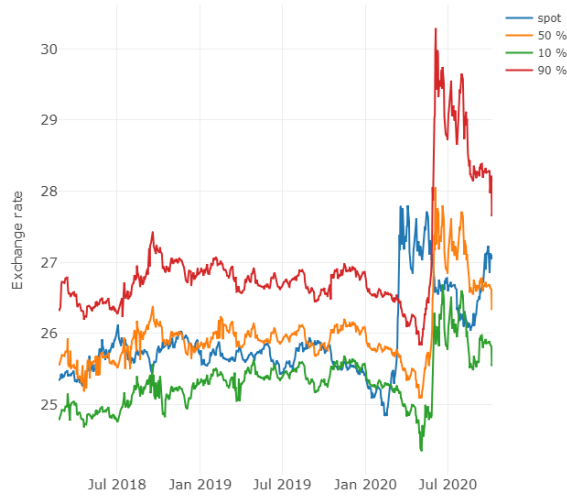
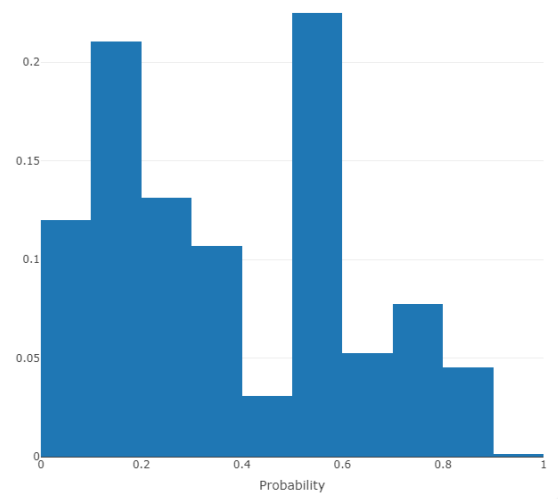


Figure 7: RND-Probability integral transform



Source: own calculation, Refinitiv

6.3 Subjective Densities

Using the methodology described above, I was able to obtain subjective densities. The development of the implied subjective and risk-neutral probability densities shows their very similar movements at the horizon of three months. However, the quantiles of subjective densities show a smaller possible range of exchange rate movements in general. In particular, sudden depreciation of the exchange rate becomes less likely; on the contrary, the probability of appreciation is mostly maintained. A significant difference between the development of subjective and risk-neutral distribution does not occur until the period of the COVID-19 pandemic in the year 2020.

Figure 9 shows that the increased range of risk-neutral probability quantiles did not fully manifest in the subjective probability quantiles. This can be illustrated by the fact that in the second quarter of 2020, 90% quantile of the risk-neutral probability quantiles almost reached 30 CZK/EUR, while at the same time, the 90% quantile of subjective probability was almost a unit less. These results are in line with the intuition about the global increase in the investors' risk aversion.

Regarding the moments of obtained densities, in the case of the subjective probability density, Figure 8 shows that the mean of the distribution, standard deviation increased, i.e., the same development as in the case of the risk-neutral distribution. Skewness slightly increased but did not reach the values in 2019.

Figure 8: Standard Deviation and Skewness

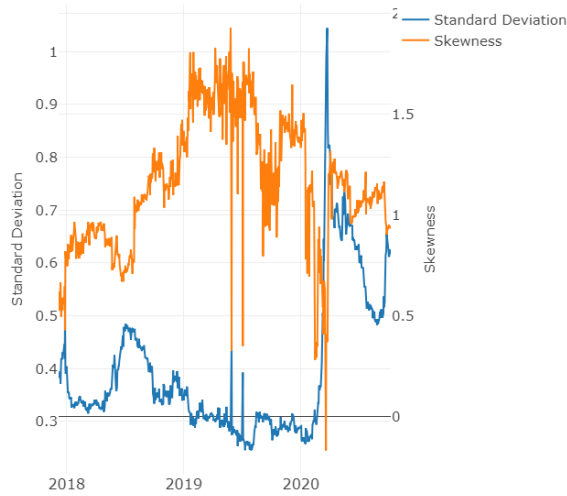
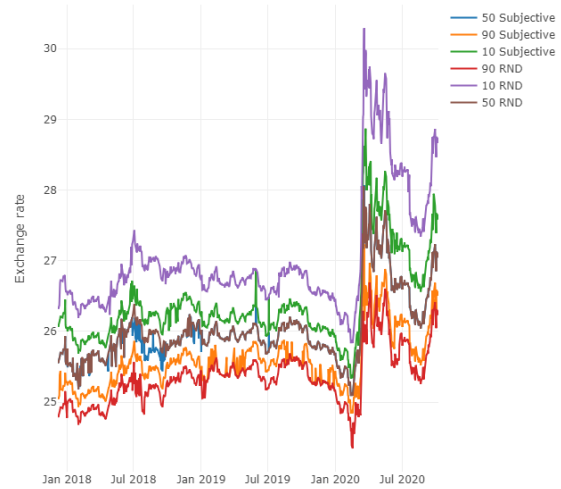


Figure 9: Comparison of risk neutral and subjective densities



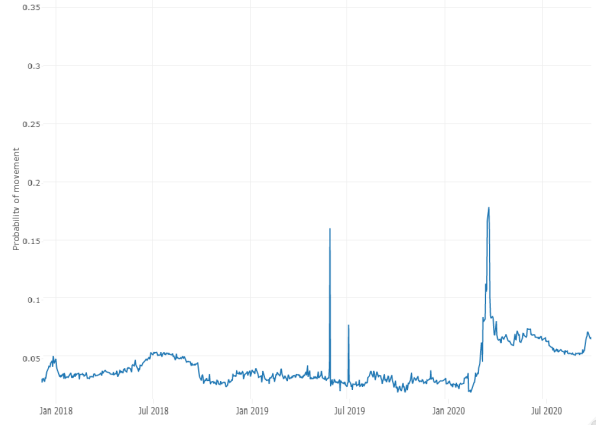
Source: own calculation, Refinitiv

Regarding uncertainty during the COVID-19 pandemic, I calculated the probabilities of extreme movement in exchange rate implied by the risk-neutral and subjective densities, as shown in Figure 10 and Figure 11. I focus on the probability of the exchange rate movement by more than 10 % in every direction compared to the spot rate. As in the previous case, I can observe that until the COVID-19 pandemic, this indicator does not show any significant shifts, and for most of the observed period, this probability is not higher than 10 %. On the contrary, this indicator reached its peak in April 2020, where the option-implied probability of a sudden movement is almost 30 %. In the case of subjective density, we can observe that for the most time during the observed period, the probability is not higher than 5 % and also reaches its peak in April 2020, where it implies a probability of sudden movement of almost 20 %. Overall, the development of the two indicators is very similar, with the difference that subjective probability densities for most of the observed period show a lower probability of a sudden and extreme movement.

Figure 10: Risk-neutral probability of movement more than 10 %



Figure 11: Subjective probability of movement more than 10 %



Source: own calculation, Refinitiv

However, it should be noted that the results need to be interpreted very carefully, as it is possible that rather than the real subjective probability of market participants, we have extracted the probability of a representative investor with a time separable utility function using Vanna Volga hedging.

However, if we assume that we have obtained real subjective probabilities of market participants, then due to the knowledge of both probability densities, we can calculate the implied risk premium rate. If we start from the equation of the price of the FX forward using risk-neutral and subjective probability measures, equating these two equations, then we are able to extract the risk premium coefficient from them. Due to the interest rate parity, the FX forward price is equal to the expected value of the exchange rate in the risk-neutral world q as well as in the real world, where the pricing kernel (risk premium) enters the valuation (Vincent-Humphreys and Noss, 2012). S_0 denotes the spot rate at the time of signing the contract, S_t denotes the spot rate at time t , r and q denote the domestic and foreign risk-free interest rate and rp denotes the risk premium.

$$S_0 = E^q S_t e^{-(r-q)t} = E^p S_t e^{-(r-q+rp)t} \quad (37)$$

$$rp = \frac{1}{t} \ln \left(\frac{E^q S_t}{E^p S_t} \right) \quad (38)$$

Results in Figure 12 show that for most of the observed period, the positive value of the risk premium with a predominantly constant size prevails from the point of view of the Czech currency. However, a notable turning point occurred in 2020, coinciding with the onset of the COVID-19 pandemic. During this period, the risk premium spiked dramatically, reaching almost 0.2%, reflecting heightened market uncertainty and a sharp response from investors. Subsequently, the risk premium began to decline, indicating that the initial exaggerated reaction by investors was gradually corrected as the situation stabilized over time. This would correspond to the perception of the Czech Republic as a small open economy and a CEE country, and

such countries are considered riskier by most investors. It is also important to realize that we observe a risk premium for the exchange rate. Therefore, it is not surprising that the obtained risk premium may be close to zero or even negative in certain periods.

Figure 12: Risk premium



Source: own calculation, Refinitiv

To evaluate the robustness and predictive power of the obtained subjective densities, the Probability Integral Transformation (PIT) was performed. However, as illustrated in Figure 13, the results do not show significant improvement compared to the risk-neutral density. On one hand, this result can be interpreted as an incorrect specification of the model, i.e., rejection of the Ross Recovery theorem. On the other hand, an alternative explanation is that the market is not able to predict future movements, even though the model is correct. To assess the robustness of the achieved results, the period of the COVID-19 pandemic was taken from the data sample, but the results remained essentially the same. (Figure 14) Formally, the Kolmogorov-Smirnov statistic was used to test the obtained distribution, where the null hypothesis about the uniformity of the obtained distribution was rejected.

Although this may seem like an overwhelming rejection, it should be noted that we are primarily concerned with a relatively illiquid market. We also focus on a relatively short period of time, especially the COVID-19 pandemic period, which can be seen as a purely exogenous shock with an extremely low probability of occurrence. Even after some time, it can be said that the reaction of monetary and fiscal authorities was unprecedented, which may have partially affected the results due to the difficulty of predicting these events. On the other hand, the

dynamics of the obtained risk premia look intuitively appealing. And partially restores the validity of the Recovery theorem.

Figure 13: Subjective Density

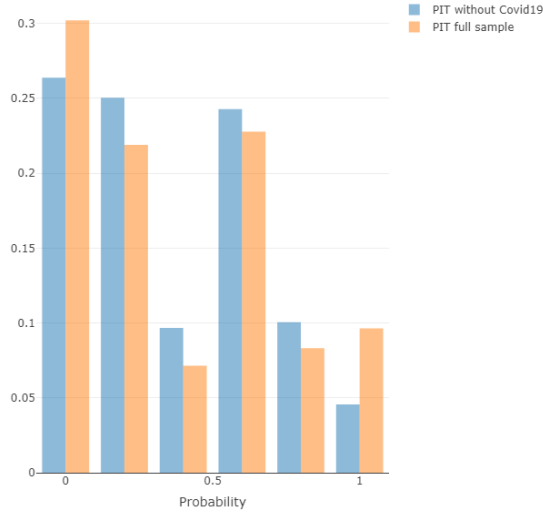
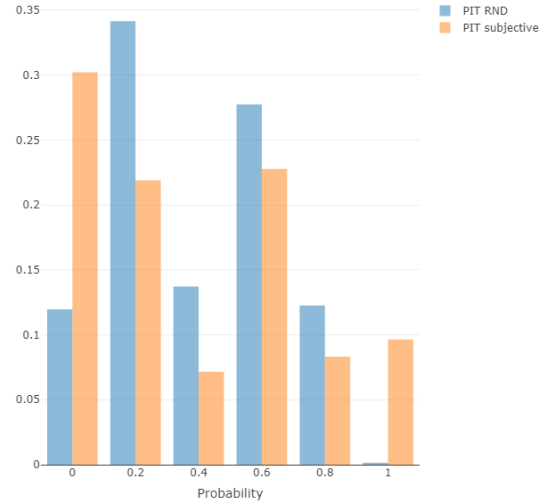


Figure 14: RND and Subjective Density



Source: own calculation, Refinitiv

Below, I show the correlation matrix of individual moments from obtained densities. Different correlations indicate additional information is found in different individual moments.

The resulting correlation matrix shows an almost perfect correlation of the first two moments of the subjective and risk-neutral distribution. On the other hand, the third and fourth moments have a correlation of only about 0.6, which suggests that new information about the subjective distribution may be found in these moments. This is also shown by the risk premium correlation, which is the highest with the subjective standard deviation and skewness.

Table 1: Correlation matrix

	Risk Premium	RND Mean	RND s.d.	RND Dkewness	RND Kurtosis	Subjective Mean	Subjective s.d.
risk premium	1						
RND mean	-0.11	1					
RND s.d.	0.66	-0.49	1				
RND skewness	0.24	0.53	-0.39	1			
RND kurtosis	-0.12	0.59	-0.66	0.81	1		
Subjective mean	-0.11	1.00	-0.49	0.53	0.59	1	
Subjective s.d.	0.73	-0.45	0.99	-0.34	-0.60	-0.45	1
Subjective skewness	0.71	0.02	0.35	0.61	0.15	0.02	0.39

7 Conclusion

In this study, risk premium, subjective, and risk-neutral densities were estimated using daily options data. Furthermore, the results of this work show that risk-neutral and subjective densities obtained using option pricing methods do not correspond to the historical realization of the real process as is tested by the probability integral transform. This can be interpreted as the market was not able to predict future developments before the COVID-19 pandemic and even lost all predictive power during the first few months of the COVID-19 pandemic. In the case of risk-neutral densities, the existence of bias in expectation is a long-known fact due to the existence of a risk premium, but if we were really able to obtain the subjective probabilities of a representative investor using the Ross recovery theorem, we have shown that the market is unable to predict future developments.

At the same time, we illustrate the time-varying risk premium of investors using data from obtained subjective and risk-neutral densities. The development of risk premium shows an approximately constant value until the period of the COVID-19 pandemic. However, after the beginning of the COVID-19 pandemic, there is a significant growth in risk premium probably due to the growth of global risk aversion. This allows an intuitive explanation of the market's inability to predict future developments as required risk premiums can change dramatically.

However, all the results presented in this study need to be interpreted with great caution because obtaining risk-neutral and subjective probabilities are two ill-posed problems that are connected to each other and are characterized by numerical instability. The fact that the obtained densities do not correspond to the realization of the processes does not necessarily mean that the market is not able to predict future developments, but the chosen method was not able to extract the subjective probabilities of market participants. On the other hand, the time-varying risk premium partially restores the validity of the recovery theorem.

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